

3(a)	<i>Mycobacterium tuberculosis</i> ; A <i>Mycobacterium bovis</i>	1
3(b)	<i>any one from</i> live in an area that has cases of TB ; recently returned from countries with TB ; born in a country with TB ; parents / grandparents whose origin country has TB ; contact with a person who has TB ; AVP ; e.g. <i>ref. to</i> compromised immune system mother who is HIV-positive	1

Question	Answer	Marks
3(c)	<i>any three from</i> decrease in incidence over time for all groups <i>or</i> decrease in incidence over time for low, lower middle, upper middle and plateau for high income group ; one example of, large / AW, differences in incidence ; e.g. between, upper middle income and, low / lower middle between high income and, low / lower middle / low and lower middle ; between upper middle and high ; decrease in incidence with income group increase (for each year), generally / overall / except for 2015 / 2016 low to lower middle ; low and lower middle similar incidence ; data / manipulated data, to support any of above points ;	3
3(d)	<i>any two from</i> <i>max 1 if no ref. to antigens</i> leprosy bacterium has similar (shaped) antigens ; memory cells, recognise / bind, antigens on leprosy bacterium ; anti-TB antibodies also bind to leprosy antigens ; AVP ; e.g. similar / same, genes so synthesise similar proteins	2

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3(e)	<i>any three from</i>	3														
	<table><tr><th>artificial active</th><th>natural passive</th></tr><tr><td>deliberate / AW A from medical staff</td><td>or not deliberate / from mother / in breast milk / across placenta ;</td></tr><tr><td>vaccine / (foreign) antigens in injection</td><td>or antibodies passed on ;</td></tr><tr><td>immune response</td><td>or no immune response ;</td></tr><tr><td>antibodies / memory cells produced</td><td>or no, antibodies / memory cells produced ;</td></tr><tr><td>longer lasting</td><td>or short-lived ;</td></tr><tr><td>protection not immediate</td><td>or immediate protection ;</td></tr></table>		artificial active	natural passive	deliberate / AW A from medical staff	or not deliberate / from mother / in breast milk / across placenta ;	vaccine / (foreign) antigens in injection	or antibodies passed on ;	immune response	or no immune response ;	antibodies / memory cells produced	or no, antibodies / memory cells produced ;	longer lasting	or short-lived ;	protection not immediate	or immediate protection ;
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